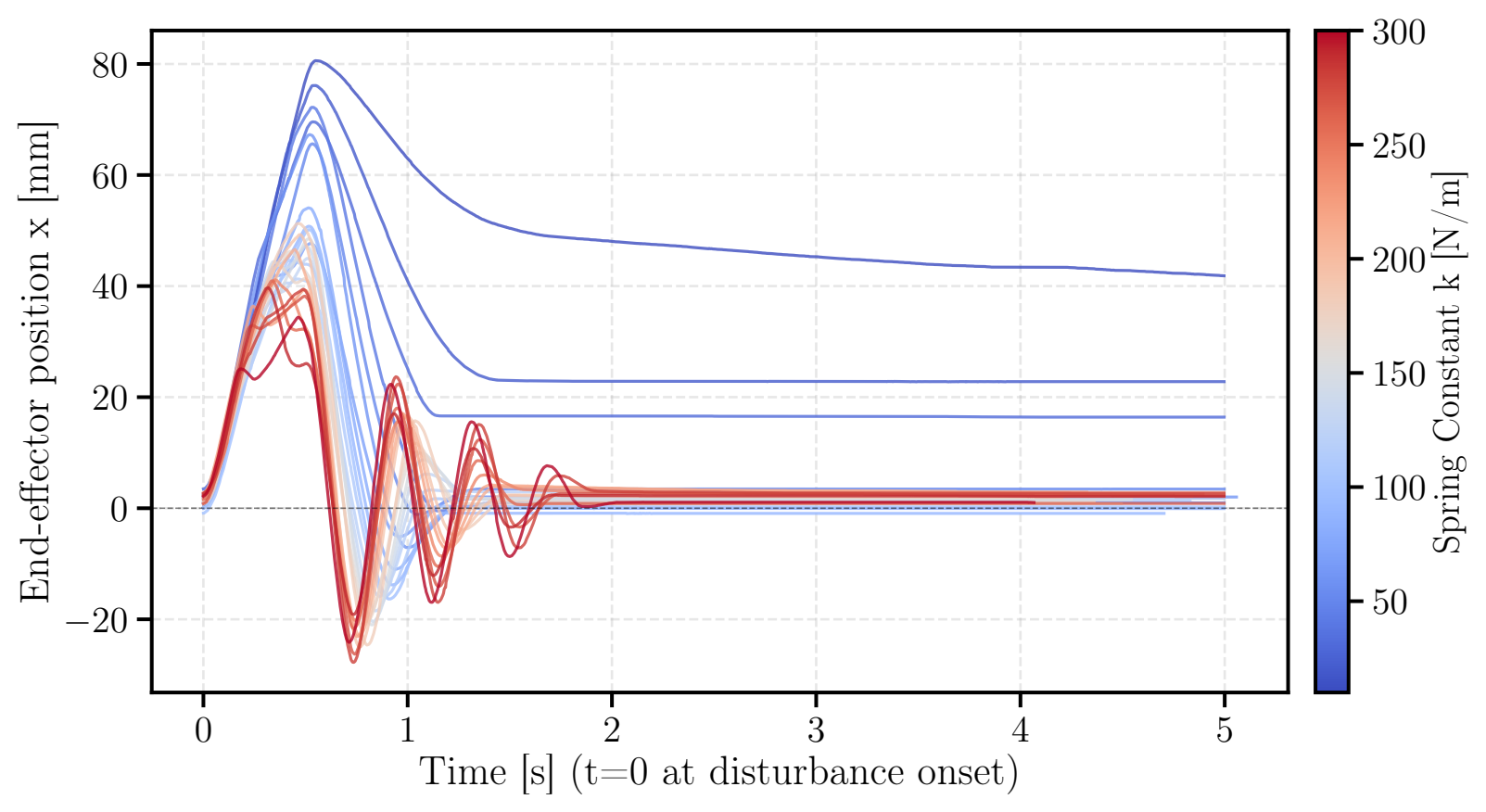

End-effector position x [mm]
80
60
40
20
0
−20
0
1
2
3
4
5
Time [s] (t=0 at disturbance onset)
Spring Constant k [N/m]
300
250
200
150
100
50